

Understanding Electric Vehicles Uptake

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Abstract: The study investigates the adoption of electric vehicles (EVs) in Washington state, focusing on the distribution of charging/fuel stations, trends in EV registrations, and regional disparities in charging infrastructure. The main aim is to understand how the availability of charging stations correlates with the population density, to analyze the trends in EV registrations, and to identify regional variations in EV adoption. This research uses multiple datasets, including electric vehicle registration transactions, charging station locations, vehicle population data by county, and population density to conduct a comprehensive analysis.

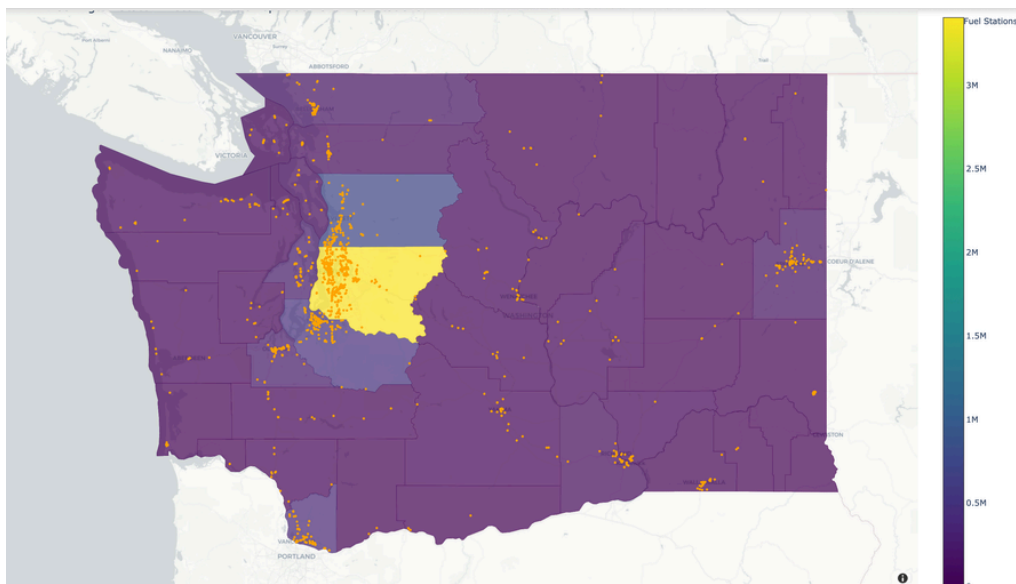
The findings reveal that charging stations are predominantly located in urban counties with higher population densities, which suggests a positive correlation between infrastructure availability and population density. Over time, EV registrations in Washington state have shown an upward trend, which indicates a growing adoption and acceptance of electric vehicles. However, there are notable disparities in registration numbers across different counties, with some counties showing much higher rates than others. These differences can be influenced by the availability of charging infrastructure, economic factors, and urbanization levels. The study concludes that to further enhance EV adoption, strategic placement of charging stations in underserved areas is crucial, alongside targeted policies to support equitable access to electric transportation.

Results:

My four original research questions were:

1. Where are the locations of electric vehicle charging stations in Washington state, and how does their distribution correlate with population density?
2. What are the trends in electric vehicle registrations over time in Washington state?
3. How do electric vehicle registration numbers vary across different counties within Washington state?
4. Are there any disparities in the availability of charging stations across different regions or communities in Washington state?

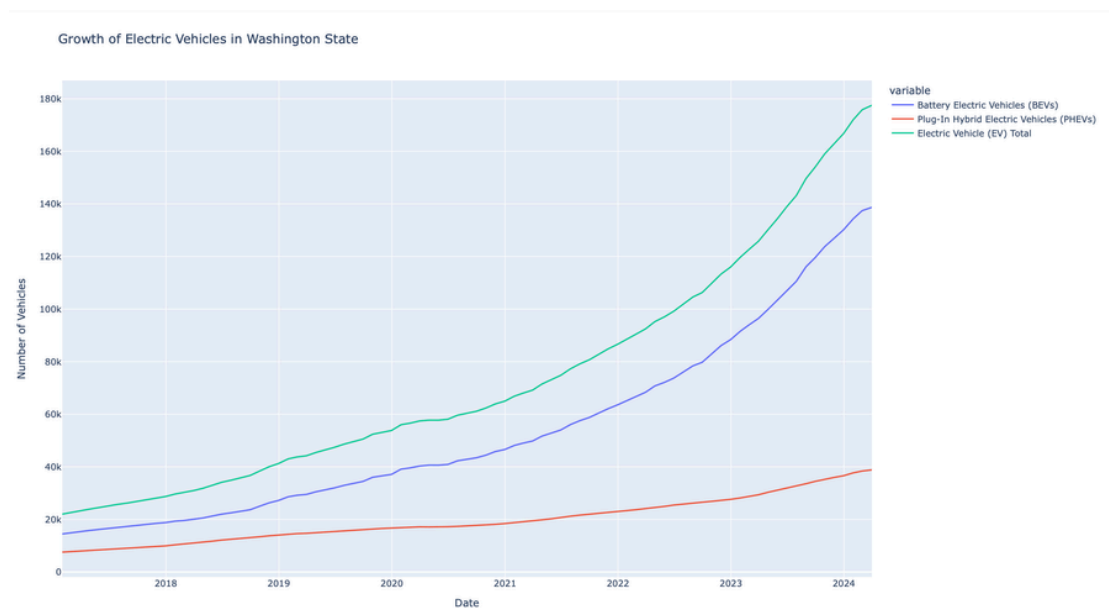
1. Geospatial Plot: Population Density between counties and location of fuel stations.



This plot shows the distribution of EV's and fuel stations across counties in Washington State. The plot uses a choropleth map to show the total number of electric vehicles per county, with lighter shades indicating higher numbers. It also overlays orange markers representing the locations of fuel stations, highlighting their geographic distribution. The purpose of this visualization is to provide a clear overview of EV adoption and fuel infrastructure across the state.

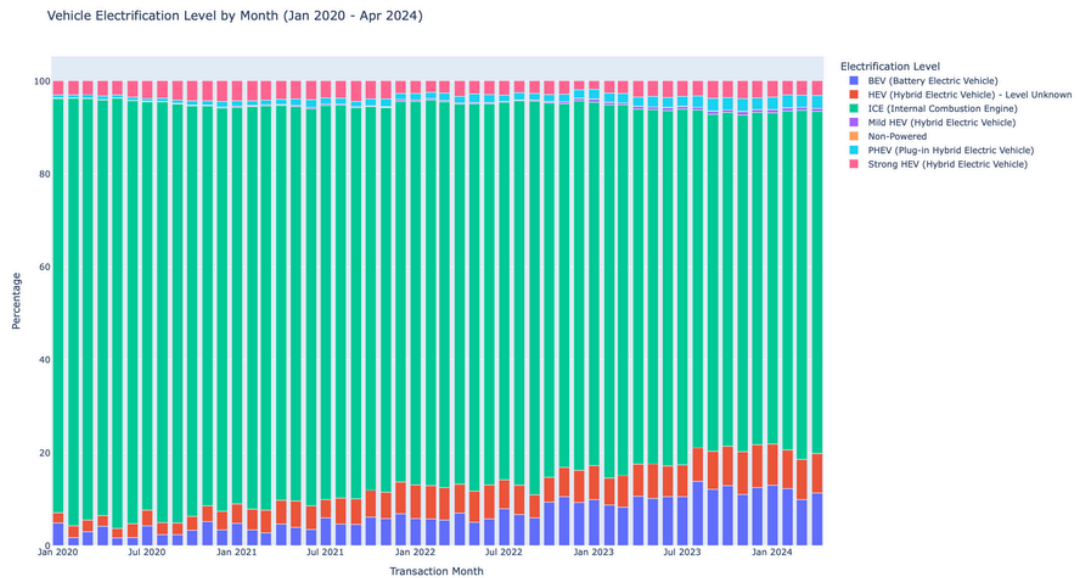
From the map, it shows that the most fuel stations are located where the most Electric Vehicles are. Snohomish, King, Skagit counties seem to have the largest population of fuel stations, and electric vehicles.

2. Line Plot: Trends in electric vehicle registrations over time in Washington state



The line plot shows the growth of EV's in Washington State from 2017 to 2024. The plot displays the growth of BEVs, PHEVs, and the total EVs. Looking at the data, it reveals a consistent upward trend in EV adoption over the years. In 2017, the total number of EVs in Washington was less than 25,000, whereas by 2024, this number had surged to slightly over 175,000, marking an impressive increase of almost 7 times. This substantial growth signifies a significant shift towards sustainable transportation in the state.

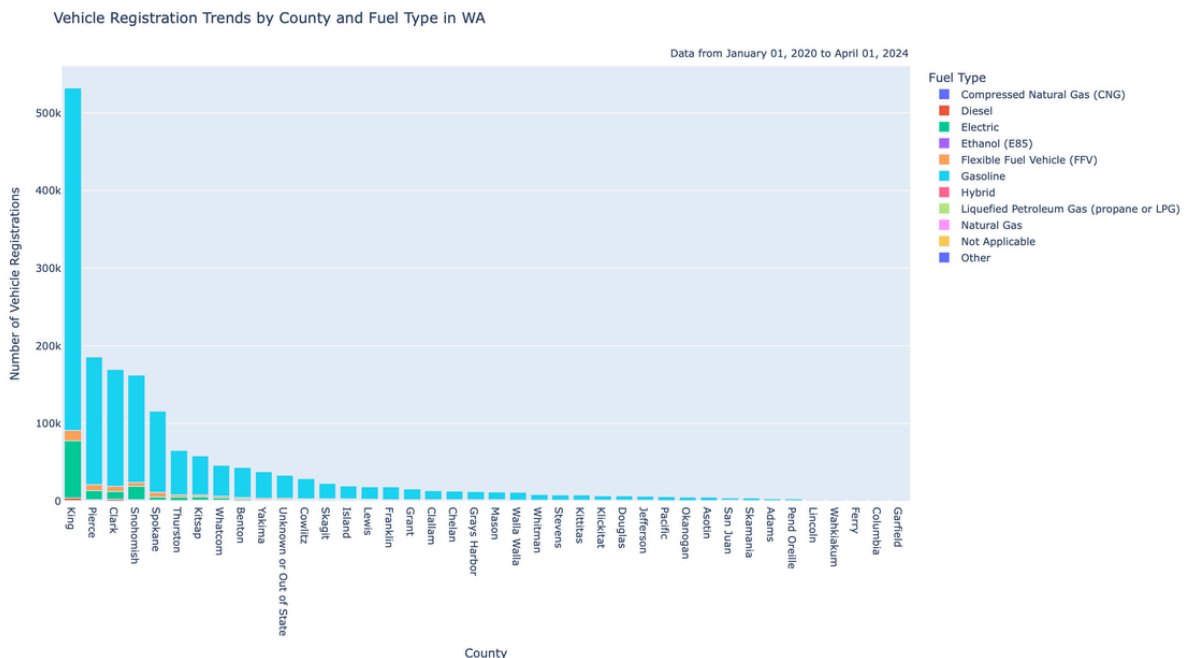
3. Vehicle Electrification Level by Month



The resulting stacked bar chart shows the percentage distribution of different electrification levels of vehicles registered each month over the specified time period. By observing the chart, we can identify trends such as increases or decreases in the registration of fully electric vehicles (Battery Electric Vehicles, BEVs), plug-in hybrid electric vehicles (PHEVs), and other categories of electrification levels.

By looking at the graph, it is clear that ICE vehicles (green), has slowly been decreasing over the past couple years. BEVs (purple), have slowly been increasing over the years. HEVs, also have been increasing over time, and PHEVs grew a little but has stayed very consistent.

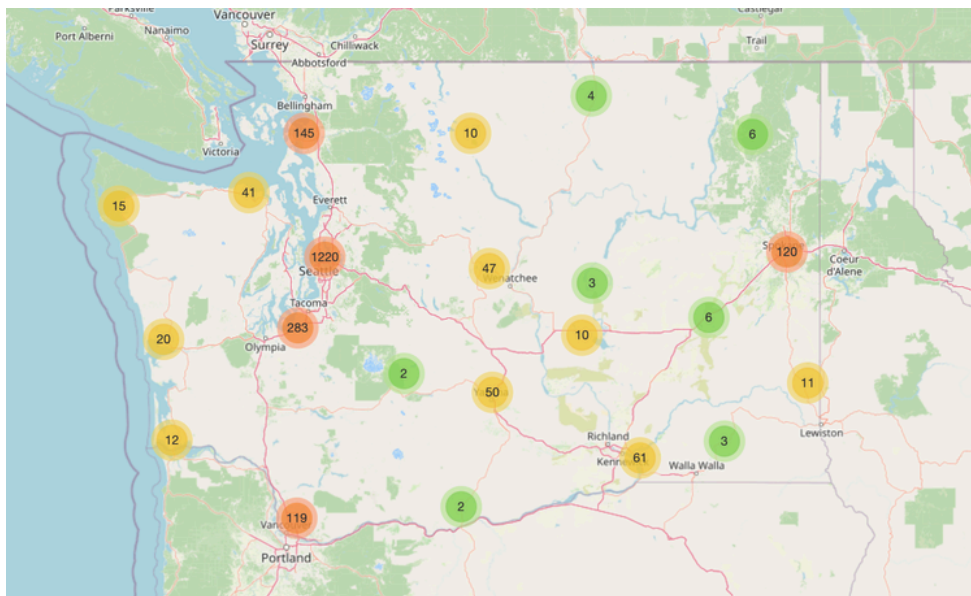
4. Vehicle Registration Trends by County and Fuel Typer



The resulting stacked bar chart displays the number of vehicle registrations by fuel type (including electric and non-electric vehicles) across different counties in Washington state. Each bar represents a county and is segmented by fuel type, showing the distribution of vehicle registrations within that county. The chart highlights how vehicle registrations, particularly for electric vehicles, differ from one county to another. This helps identify which counties have higher or lower adoption rates of electric vehicles.

From the chart, it is clear how King County has the greatest amount of electric car registrations compared to other counties, as well as have the most registered cars since 2020. Most registered cars are gasoline fuel types in King County, there is a little less than 100k electric fuel type cars.

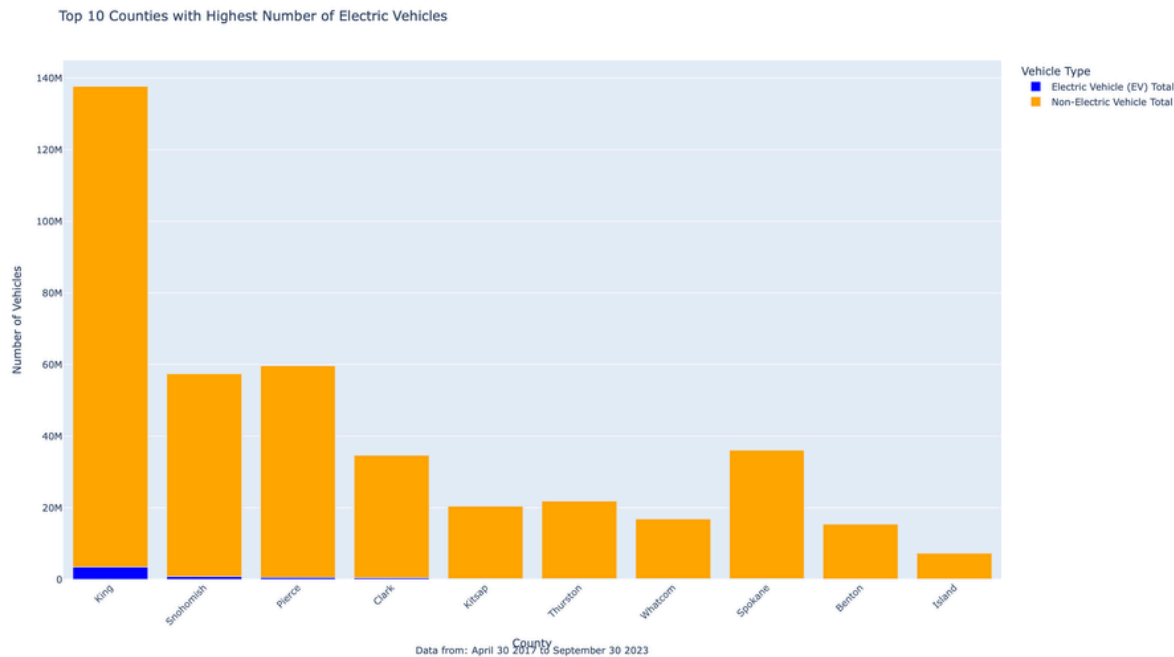
5. Fuel Stations across Washington



The interactive map displays the locations of fuel stations across Washington state, with a specific focus on electric vehicle (EV) charging stations. Each marker on the map represents a fuel station, with color coding to differentiate between electric (blue), compressed natural gas (green), and other types of fuel (red). The map provides a clear visual representation of where EV charging stations are located within Washington state. This helps identify areas with high or low densities of charging infrastructure.

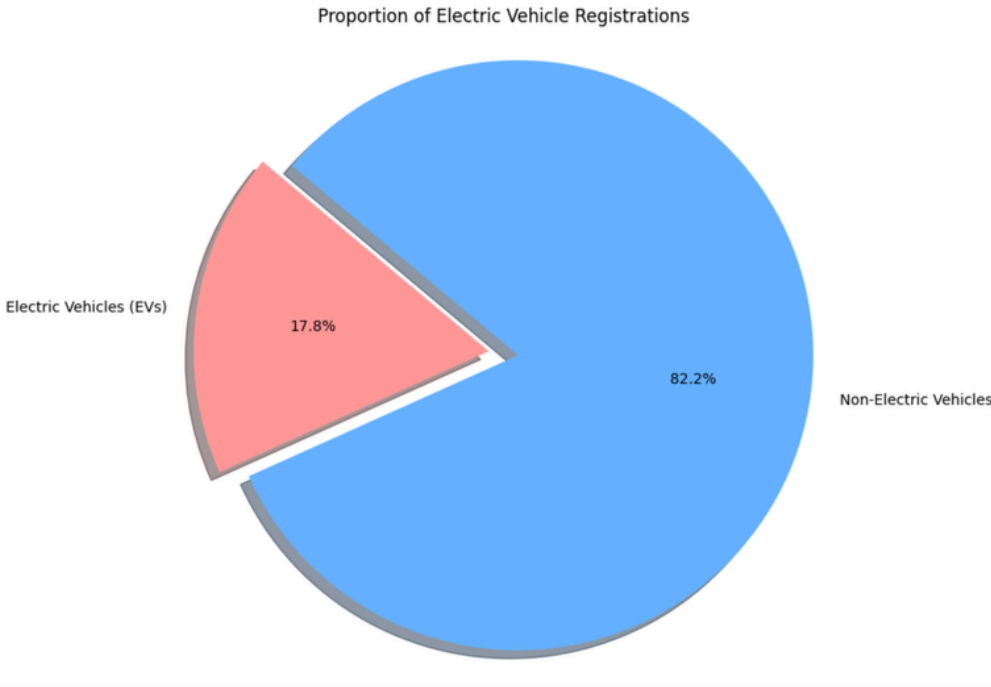
The orange colored circles on the map are where the most stations are located. From this map, we can see that near King county, Snohomish County, Whatcom County, and Spokane County have the most fuel stations.

6. 10 Counties with Highest number of Electric Vehicles



The stacked bar plot illustrates the top 10 counties in Washington State with the highest number of electric vehicles (EVs). Each bar represents the total number of vehicles, with the blue segment indicating EVs and the orange segment representing non-electric vehicles. King County stands out as the county with the highest EV adoption, followed by Snohomish and Pierce counties. The data range from the earliest available date to the latest recorded date, reflecting the increasing popularity of EVs across these counties over time.

6. Proportion of EV Registrations



The resulting pie chart shows the proportion of electric vehicles (EVs) versus non-electric vehicles registered in Washington state. The chart provides a clear visual representation of the share of EVs within the total vehicle registration count.

Data Biases:

The only bias in the data is that most of my datasets only had data on Washington State

Conclusion:

This study explores the adoption landscape of electric vehicles (EVs) in Washington state, looking at important aspects such as the location of charging stations, EV registration trends, and regional differences in the infrastructure for charging EVs. Several important conclusions are drawn from a thorough examination of numerous datasets, including EV registration transactions, charging station locations, and population density.

First, the geospatial plot shows how EVs and fuel stations are distributed among Washington State counties. The map indicates a relationship between the amount of fuel stations and the number of electric vehicles (EVs), with urban counties such as Snohomish, King, and Skagit having the greatest EV populations. This implies that there is a positive correlation between infrastructure availability and population density.

Furthermore, the analysis of EV registration trends over time shows a consistent upward trajectory in adoption, signifying a notable shift towards sustainable transportation. Between 2017 and 2024, the total number of EVs in Washington surged from less than 25,000 to over 175,000, which is a substantial increase of nearly sevenfold. This growth underscores the increasing acceptance of EVs.

Potential difficulties are highlighted by differences in EV adoption and charging infrastructure amongst counties. With the greatest number of registrations and most number of electric vehicles, King County stands out as the leader in the adoption of EVs. However, there are differences in other counties with relatively lower adoption rates. These differences could be caused from various factors such as economic conditions, urbanization levels, and the availability of charging infrastructure.

In order to further promote EV adoption, the research emphasizes how crucial it is to properly locate charging stations, especially in underserved areas. Furthermore, for widespread adoption to occur in all communities, specific regulations that support fair access to electric vehicles are necessary.

Further Research:

As I was working on this project, one of the other questions I wanted to explore is how socioeconomic status would affect EV ownership. If I had the necessary data, it would have been useful to have plots to show the relation between them. Another question I wanted to address was the environmental impacts of EVs. I did not have the data or the time to address this question.

Challenge Goals:**1. New Libraries**

I made it one of my challenge goals to incorporate a new library that we hadn't discussed in class. I choose to use Plotly, a visualization tool that provides interactive charts. I made a number of visualizations for my project using Plotly, including stacked bar charts, line plots, and choropleth maps. I was able to create visually appealing and informative plots with Plotly, which improved how my study findings were presented. For instance, I plotted the rise in Washington state's EV registrations over time using Plotly to illustrate the adoption trend of EVs from 2017 to 2024. Additionally, I used Folium, another library that was new to me, to create an interactive map showing the locations of fuel stations across Washington state, with a specific focus on EV charging stations. Folium allowed me to overlay markers on a map and customize their appearance based on different attributes, such as fuel type.

One of the challenges I encountered when learning to plot using Plotly and Folium was understanding the syntax and functionality of these libraries. Initially, I found the syntax of Plotly to be quite different from what I was used to. Plotly operates on the concept of "figures" and "traces," which was different. With Folium, I found it challenging to navigate its documentation and understand the various parameters and options available for customizing markers, popups, and map styles

2. Multiple Datasets

I made it another one of my challenge goals to join multiple datasets. I choose this challenge because I was already working with a 5 datasets, and I needed to merge/join them together for my plots. What I was trying to achieve to create a dataset where there would be number of registration of EVs, with the population density. This dataset would give the ratio to see the amount of EV's registered within the population density of each county. Similarly I wanted to get the number of EVs, by county within the population density. I also wanted to figure out the number of fuel stations within each county.

The first two goals were harder to achieve than I thought. This was because the data was harder to clense because all my data had millions of rows. I tried to come up with a unified dataset so it would be easier to project in our code. The volume of the data was the biggest challenge.

References:

- <https://data.wa.gov/Transportation/Electric-Vehicle-Share-of-New-Registrations/wzin-vviu>
- https://afdc.energy.gov/fuels/electricity_locations.html#/analyze
- https://data.wa.gov/Transportation/Electric-Vehicle-Population-Size-History-By-County/3d5d-sdqb/data_preview
- https://public.opendatasoft.com/explore/dataset/us-county-boundaries/table/?flg=en-us&disjunctive.statefp&disjunctive.countyfp&disjunctive.name&disjunctive.namesad&disjunctive.stusab&disjunctive.state_name
- <https://www.census.gov/data/datasets/time-series/demo/popest/2020s-counties-total.html>
- <https://plotly.com/python/>
- <https://www.geeksforgeeks.org/python-plotting-google-map-using-folium-package/>
- <https://stackoverflow.com/questions/62517929/python-folium-map-developement>
- <https://idp.mrstride.com/final-project/handouts/final.html>
- <https://www.geeksforgeeks.org/python-plotly-tutorial/>